

Hyper-raising and the subject position in Brazilian Portuguese

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1. Introduction

In this paper I propose that Brazilian Portuguese (BP) subjects occupy a mixed A/A' position between Spec,CP and Spec,TP (i.e., Spec,A/A'P) and argue that hyper-raising constructions are derived via movement to this position as depicted in (1).

(1) [CP [A/A'P DP_i [TP ... [CP [A/A'P t_i [TP ...

I also claim the subject is not frozen for movement in the embedded finite clause because feature checking there involves the A/A' projection, and not a pure A or A' projection. I am using "hyper-raising" (HR) in the sense of Ura (1994), thus referring to "languages that allow raising from the **subject** position of a tensed (or finite) clause" (p.65; emphasis added by the author). Consider (2).

- (2) a. Os meninos parecem que *t* comeram maçã.
 the boys seem.3PL that ate.3PL apple
 'It seems that the boys have eaten apples'
- b. *pro*_{EXPL} Parece que os meninos comeram maçã.
 seem.3SG that the boys ate.3PL apple
 'It seems that the boys have eaten apples'

Notice that the 3rd person plural DP subject *os meninos* ('the boys') triggers agreement with both verbs in the embedded and in the matrix clause (2a). When movement doesn't take place, an expletive *pro* is inserted instead, satisfying matrix EPP and triggering default (3SG) agreement (2b). HR constructions like (2a) are challenging since subject raising out of finite clauses is expected to violate the Activation Condition (Chomsky 2000, 2001), as movement from a Case-checking position to another Case-checking position is arguably banned (cf. **The boys seem (that) are happy*). In other words, DPs that have entered a Case-checking relation should be frozen in place and no longer move. Another potential issue is improper movement, as movement through the embedded Spec,CP to the matrix position would qualify as movement from an A' position to an A position (e.g., from embedded Spec,CP to matrix Spec,vP). In the following, I will argue BP subjects occupy a mixed A/A' position which allows HR constructions to be derived without inducing freezing effects.

The proposal that subjects in HR languages occupy Spec,A/A'P straightforwardly accounts for the mixed A/A' properties of BP subjects and will also account for the unavailability of a subject-object inverse scope reading in the language. I also claim that movement out of A/A'P voids the freezing effect of feature checking. Such movement recreates the configuration needed for agreement in the matrix clause, which makes the same DP trigger agreement both in the embedded and in the matrix clause.

2. The data: hyper-raising in Brazilian Portuguese

One could wonder whether the trace in (2a) isn't actually *pro*. In this section I will show it is not, as has been argued in the literature. First, BP does not allow referential 3rd person null subjects, which would make the licensing of *pro* in embedded clauses unexpected.

- (3) *Embarcou no trem .
 boarded.3SG in.the train
 'He/She boarded the train'

Rodrigues (2004:03)

Also, 3rd person null subjects in embedded contexts are not free in reference, being dependent on the matrix DP subject.

- (4) Ele₁ disse que e_{1/*2} embarcou no trem.
 He said that boarded in.the train
 'He said that he boarded the train'

Rodrigues (2004:03)

As observed by Duarte (1996), BP null subjects in embedded finite clauses do not obey the *Avoid Pronoun Principle*¹ (APP), thus allowing overt non-emphatic subject pronouns instead of *pro* (5). The null subject in BP alternates freely with the overt pronoun, which suggests that the former is not *pro*, given that *pro* cannot be involved in such free variation (cf. Montalbetti's 1984 Overt Pronoun Constraint).

- (5) O João disse que (e_{1/2}) gosta da Maria.
 the João said that he likes of.the Maria
 'João said that he likes Maria'

Rodrigues (2004:79)

Notice that the null subject freely alternates with the overt pronoun, which shows that the latter is not being "avoided" in any sense. The silence in (5) being a trace is fully compatible with the APP. Additional evidence that the matrix subject is derived via movement can be seen in (6), where a null subject is prohibited within the relative clause.

- (6) a. Parece que [o bolo que o João comeu não tava bom.]
 seems that the cake that the João ate not was good
 'It seems that the cake that João ate wasn't good'

- b. *O João parece que [o bolo que t comeu não tava bom.]
 the João seems that the cake that ate not was good

Nunes (2008:10)

Also, reconstruction effects are attested for HR constructions. In (7) we see that subject DPs of sentential idioms can hyper-raise.

- (7) a. A vaca foi pro brejo.
 the cow went to.the swamp
 'Things went bad.'

¹"Empty categories have preference over overt pronouns" (Chomsky 1981:65).

- b. A vaca parece que foi pro brejo.
 the cow seems that went to.the swamp
 'It seems that things went bad.'

Finally, in (8) the matrix DP *ninguém* ('nobody') licenses the embedded Negative Polarity Item (NPI) *um dedo* ('a finger').

- (8) a. Ninguém mexeu um dedo pra me ajudar.
 nobody moved a finger to me help
 'Nobody lifted a finger to help me'
- b. *Ninguém disse [que a Maria mexeu um dedo pra me ajudar.]
 nobody said that the Maria moved a finger to me help
 'Nobody said that Maria didn't lift a finger to help me'
- c. Ninguém parecia [que t ia mexer um dedo pra me ajudar.]
 nobody seemed that went move a finger to me help
 'It seemed that nobody was going to lift a finger to help me' Nunes (2008:11)

Notice that the null subject in (8c) being a trace is compatible with the requirement that the NPI licenser and the NPI must be clause mates. Therefore, I take null subjects of embedded clauses of HR constructions to be traces (i.e., deleted copies) of movement, in line with the available literature.

3. Previous accounts to hyper-raising in Brazilian Portuguese

3.1. The ϕ -incomplete finite T approach to hyper-raising

As observed in Nunes (2008:86), Ferreira (2000, 2004, 2009) and Rodrigues (2002, 2004) have argued that with the weakening of the verbal paradigm in BP, the finite T in the language ceased to license referential *pro*, and referential null subjects were reanalyzed as A-traces. Ferreira (2000, 2004, 2009) suggests that BP finite Ts are ϕ -ambiguous, that is, they could either be associated with a complete or with an incomplete set of ϕ -features. In this approach, a ϕ -complete T would freeze the subject in the embedded clause by assigning structural nominative Case to it, deriving (2b). A ϕ -incomplete T, on the other hand, would fail to assign nominative Case, which would make the embedded subject eligible to move into the matrix clause, thus deriving (2a). This is under the assumption by Chomsky (2000, 2001) that nominative Case is assigned as a side effect of ϕ -complete agreement between T and the subject, which correctly predicts the impossibility of movement in **John seems (that) is tired*, but the need for movement in *John seems to be tired*.

Nunes (2008) implements this idea by suggesting that (i) person and number features can be combined either in the numeration or in the morphological component, which would allow a "personless" (ϕ -incomplete) T in syntax, and that (ii) nominative pronouns in BP are underspecified (cf. Nunes 2020), which would also account for the "weakening" in BP verbal inflection. The motivation for underspecification in the pronominal system refers to the loss of verbal inflection in BP, where 2nd singular, 3rd singular, and 1st plural pronouns all trigger default agreement (3rd singular) on the verb (9a). 2nd plural and 3rd plural pronouns, on the other hand, trigger 3rd plural agreement (9b).

- (9) a. *Você/ Ele/Ela/A gente* *canta*
 you.SG/ he/ she/ we *sing.PRES.IND.3SG*
 'You/He/She/We sing(s)'
- b. *Vocês/ Eles/ Elas* *canta-m*
 you.PL/ they.MASC/ they.FEM *sing-PRES.IND.3PL*
 'You/They sing'

He accounts for (9) by claiming the pronouns in (9a) are radically underspecified for person and number, and the ones in (9b) have only a number feature (i.e, plural). The pronouns *eu* ('I') and *nós* ('we'), on the other hand, would be specified with only one feature, which he claims is enough to identify them: the former is specified as [Person.Number: SG], and the latter as [Person.Number: 1]. He shows that you only need the singular value to identify the 1st person singular pronoun morphologically, since there is no person distinction between the singular forms *você*, *ele*, *ela* and *a gente*. In the case of *nós*, however, we only need the person value "1", since the plural forms *vocês*, *eles* and *elas* have no person distinction morphologically expressed (see 10; remember that BP uses 3rd person as default).

- (10) *Eu cant-o /* *Nós* *canta-mos*
 I sing-PRES.IND.1SG / *we* *sing-PRES.IND.1PL*
 'I sing/'We sing'

It is not at all clear to me how Brazilian Portuguese nominative pronouns will end up being properly interpreted if they are "maximally underspecified" (Nunes 2020:4659). If the pronouns *você* (2sg), *ele* ('he'), *ela* ('she'), and *a gente* ('we') are equally unspecified for person and number, where the different interpretations that we obviously still have among them would be encoded? How is the semantic component going to distinguish the 2nd person singular from the 3rd person singular if they are identical objects in the derivation? The fact the morphological component is able to properly realize the agreement inflections without needing to access a full specification of person and number features is not itself evidence of underspecification of the pronominal system. Syncretism across "cells" in agreement patterns is a phenomenon widely attested cross linguistically, and there is plenty of work within the Distributed Morphology framework, for instance, that deals with such cases as underspecification in the phonological exponents, but not in the syntactic nodes, which are generally assumed to be fully specified.² Therefore, I don't see any reason to interpret the loss of inflection markers in BP as the aftermath of Agree with underspecified pronouns.

On the possible different timings of insertion of person and number features (either in the numeration or in the morphological component), which would allow a "personless" ($\varnothing$ -incomplete) T in syntax, no supporting evidence was presented, and I couldn't think of a way to test this claim, as T would always surface the same. Also, the only alleged "visible" effect of the proposed timing difference seems to be DP movement out of the finite domain, which is precisely the phenomenon the analysis aims to account for.

²Cf. subset principle: Harley & Noyer (1999:04). Underspecified vocabulary items compete for insertion into fully specified syntactic nodes. See also Sauerland's (1995) account of the syncretic pattern of Dutch strong adjectives desinences, which crucially relies on the subset principle.

3.2. The A features in C approach to hyper-raising

Fong (2018, 2019) shows that accusative subjects of embedded clauses in Mongolian are Case-marked by the matrix verb (and can therefore hyper-raise).

- (11) a. Bat chang-aar [Dorj sain seheetin gej] khel-sen.
 Bat loud-INST Dorj.NOM good noble COMP say-PST
 'Bat said loudly that Dorj is good and noble'
 b. Bat chang-aar Dorj [ec sain seheetin gej] khel-sen.
 Bat loud-INST Dorj.NOM good noble COMP say-PST
 'Bat said loudly that Dorj is good and noble' Fong (2018:02)

As already mentioned, Ferreira (2000, 2004, 2009) and Rodrigues (2002, 2004) show that embedded null subjects are traces of movement in BP sentences like (12), where the DP moves to a θ -position.

- (12) O João disse [t que comprou um carro].
 the João said that bought a car
 'João said that he bought a car' Ferreira (2009:29)

Now let's consider (13).

- (13) a. A Maria me viu (estudando).
 the Maria 1sg.ACC saw studying
 'Maria saw me (as I was studying)'
 b. Eu vi que t estou estudando pouco.
 I saw that am studying little
 'I realized I'm not studying much'

(13a) shows the verb *ver* ('see') assigns accusative Case. (13b) shows a construction where the embedded subject moves to a θ -position out of a finite clause, as in (12). Now look at (14).

- (14) *Me vi que t estou estudando pouco
 1sg.ACC saw that am studying little
 (Int.) 'I realized I'm not studying much'

Given that BP embedded finite subjects do indeed move to a θ -position in the matrix, I don't see how (14) would be excluded in Fong's approach, since we would expect the matrix verb to assign accusative to the embedded subject, which would therefore be able to raise as it happens in Mongolian's (11). Let's say that whatever mechanism typically assigns abstract Case in BP would be unavailable in the embedded clause in (12). Under Fong's analysis, the matrix verb, which is an accusative Case assigner, should be able to Case-mark the embedded subject as it does in (11b).

The analysis developed here differs from Fong (2018) in the following. She suggests HR in BP is derived by subject movement to Spec,CP triggered by A-features in C, which create A-positions. For her, HR is an instance of A-movement even though it passes through Spec,CP. Here, HR will not be derived via movement to Spec,CP, but to a lower position, Spec,A/A'P.

3.3. C as a composite A/A' probe

Lohninger, Kovač & Wurmbrand (2022) propose that the matrix predicate in HR constructions as in BP lexically selects a special CP, i.e., "CP.R", which "bundles the A-properties of a predication R[elator]P[hrase] [...] with the A'-properties of a (regular) CP" (p.12).³ They claim that the mixed A/A' nature of this projection allows the DP in its Spec to be targeted by matrix A-probes without involving improper A-after-A' dependencies. Notice, however, that our analyses are not notational variants. The CP.R projection is restricted to HR constructions, and it is not a (locally moved) subject-dedicated position as Bošković's (2021, in press) A/A'P, which is the projection I will rely on in the analysis to be developed here. Our approaches also make different predictions. If Fong (2018) is correct in her proposal that a composite A/A' probe in C is responsible for deriving hyper-raising to **object** in Kipsigis (15) (p.13-4),⁴ then it's not clear to me how we would be able to prevent HR to object in BP under the CP.R approach, which is ungrammatical (16).

- (15) mócè pè:ndó Mù:sá [kó-tíl Kíplàjàt].
wants meat Musa 3S.SUB-cut Kiplangat
'Musa wants that Kiplangat cut the meat.'

Jake & Odden, 1979 (11)

- (16) *O bolo parece que o João comeu.
the cake seems that the João ate
'The cake seems that John ate.'⁵

4. A/A'P

Bošković (2021, in press) has independently argued for the existence of a subject-dedicated position above TP and below CP, claiming it is an instance of a mixed A/A' position. He shows this position hosts local *wh*-subjects, which move straight to this position without passing through Spec,TP. This projection doesn't host *wh*-objects, embedded (long-distance moved) *wh*-arguments, and *wh*-adjuncts, which all move to Spec,CP instead. As an A/A' position, it can (i) satisfy EPP,⁶ (ii) check nominative Case, (iii) check the [+WH] feature and, I will argue, trigger agreement.

He shows that *who* in *who left?* is neither in Spec,TP nor in Spec,CP, but it must be in between. West Ulster English allows quantifier float under *wh*-movement (cf. 17). One argument that *who* is not in Spec,TP is that a subject in Spec,TP is unable to float a quantifier in the θ -position in passive (17b). If *who* in (17a) were to pass through Spec,TP (or if it were located in Spec,TP), the quantifier would be floated from the same position in both (17a) and

³Cf. also Wurmbrand (2019a, b).

⁴"An appropriate goal for this type of hybrid probe has to have matching features. If the embedded subject does not have them, it will be skipped over them. Conversely, if the object bear A/A'-features, it will be an adequate goal, allowing it to be hyper-raised." (Fong 2018:14).

⁵A focused or topicalized object is allowed, but it would require a comma intonation separating the fronted object with the rest of the clause. This is of course not the relevant derivation here.

⁶In Bošković's proposal, the A/A'P is the highest projection in the EPP domain, which corresponds to a return to split IP. In his system, EPP is checked in the highest projection of the EPP domain, i.e., Spec,A/A'P. Such domain looks like (i). For instance, English regular subjects would be in Spec,AgrsP, quirky subjects are good candidates for Spec,TP.

(i) [A/A'P [AgrsP [TP ...

(17b), so the former should be ungrammatical. Therefore, the *wh*-subject in (17a) moves above TP, which means it must be able to satisfy the EPP and get nominative Case above TP.

(17) a. Who [_{TP} (**t*) was arrested all *t* in Duke Street?]

b. *[_{TP} They were arrested all *t* last night.]

c. What did [_{TP} he say all *t* that he wanted?]

Bošković (2021:03)

However, *who* and *what* in (17) cannot occupy the same position in the left periphery. If that were the case, the contrast in (18) would be unexpected. Notice that the *wh*-object moves above the topic, a position unavailable for *wh*-subjects, suggesting the former is higher than the latter.

(18) a. ?Mary wonders which book, for Kim, Peter should buy.

b. *Mary wonders which student, for Kim, should buy that book. Bošković (2021)

Another interesting evidence that *wh*-subjects and *wh*-objects are in different positions comes from Igbo, a language spoken in southeastern Nigeria, where they are marked differently (19). Notice that the focus particle is obligatory with the former and prohibited with the latter.

(19) a. Ònyé *(ka) Òbí hùrù n'-áhíá?

who FOC Obi saw P-market

'Who did Obi see at the market?'

b. Ònyé (*ka) hùrù Ádá n'-áhíá?

who FOC saw Ada P-market

'Who saw Ada at the market?'

Amaechi and Georgi (2019)

Further, the fact (20a) allows for an inverse scope reading shows that the object *everyone* scopes over a quantifier in Spec,TP, the regular subject position. Then, with the impossibility of inverse scope in (20b), he concludes *who* is not located in Spec,TP (cf. also Mizuguchi 2014), and it must be in Spec,A/A'P, as Spec,CP is also excluded (cf. 18).

(20) a. Someone loves everyone. ($\exists > \forall$; $\forall > \exists$)

b. Who loves everyone? ($\exists > \forall$; $*\forall > \exists$)

Bošković (2021)

The A/A' position is not confined to *wh*-subjects. Bošković (2022) shows it can also host V2 subjects, *Only*-subjects, Imperative subjects, Locative inversion, Singlish non-agreeing subjects and Defaka focus subjects. Spanish (regular) subjects, Chinese subjects and quirky non-subjects are also good candidates for that position. In the next section I will argue that both *wh* and regular Brazilian Portuguese subjects occupy Spec,A/A'P. The mixed A/A' proposal captures the contrast we see between (21a) and (21b), in which extraction across *who* in the former is less degraded than extraction across *how* in the latter (cf. Chomsky 1986 where this contrast is reported). The account is given in (22).

(21) a. ?What_i do you wonder who bought *t_i*?

b. ??What_i do you wonder how she bought *t_i*?

(Bošković 2022)

(22) a. ?[_{CP} what do you wonder [_{A/A'P} who bought *t*]]

b. ??[_{CP} what do you wonder [_{CP} how she bought *t*]]

5. A/A'P in Brazilian Portuguese

Local BP *wh*-subjects occupy Spec,A/A'P, as in English. Notice that BP displays the same topicalization pattern observed in (18), which indicates *wh*-objects are higher than *wh*-subjects (23).

- (23) a. Maria quer saber que livro, pro Pedro, o João comprou.
 Maria wants to.know which book to.the Pedro the João bought
 'Maria wants to know which book, for Pedro, João has bought.'
 b. *Maria quer saber que aluno, pro Pedro, comprou o livro.
 Maria wants to.know which student to.the Pedro bought the book
 (Int.) 'Maria wants to know which student, for Pedro, has bought the book.'

Another evidence for the availability of A/A'P in BP is the ellipsis data below. *Quem nunca* ellipsis is only available with a locally moved *wh*-subject (24b, 25b). With *wh*-objects (25c), long distance (LD) *wh*-arguments, including subjects, (25c,25d), and *wh*-adjuncts (26), this ellipsis is disallowed. The sentences in (a) correspond to the discourse antecedent of the elliptical constructions in (b-d). The non-elided versions of the ellipses being examined are all fully acceptable.

- (24) a. Pedro beijou João.
 Pedro kissed João
 b. Quem nunca [~~t~~beijou João]?
 who never kissed João?
 'Who has never kissed João?'
 c. *Quem Pedro nunca [~~t~~beijou t]?
 who Pedro never kissed
 'Who has Pedro never kissed?'
 (25) a. Maria disse que Pedro beijou João.
 Maria said that Pedro kissed João
 b. Quem nunca [~~t~~disse que Pedro beijou João]?
 who never said that Pedro kissed João
 'Who has never said Pedro kissed João?'
 c. *Quem Maria nunca [~~t~~disse que Pedro beijou t]?
 who Maria never said that Pedro kissed
 'Who has Maria never said Pedro kissed?'
 d. *Quem Maria nunca [~~t~~disse que t beijou João]?
 who Maria never said that kissed João
 'Who has Maria never said kissed João?'
 (26) a. Pedro beijou João na festa.
 Pedro kissed João at.the party
 b. *Onde Pedro nunca [t beijou João]?
 where Pedro never kissed João
 'Where did Pedro never kissed João?'

Only local *wh*-subjects are allowed in *Quem nunca* ellipsis constructions, whereas non-local *wh*-subjects (i.e., *wh*-objects, LD *wh*-arguments, *wh*-adjuncts) are not. This is exactly the cut predicted under the subjects-in-Spec,A/A'P analysis. By assuming with Lobeck 1995 that [+WH] Spec-head agreement is a necessary condition for ellipsis licensing in cases of sluicing, I propose that *nunca* heading the A/A' projection licenses ellipsis when it stands in a Spec-head relation with *Quem*_[+WH], which is only possible when *quem* corresponds to the local subject. Also notice that the *wh*-elements in (24b) and (25b) cannot correspond to the object. Check the table below for a summary.

(27) *Quem nunca* ellipsis

	<i>Quem+nunca</i> XP	e.g.,
local <i>wh</i> -Subj	✓	24b,25b
local <i>wh</i> -Obj	*	24c
LD <i>wh</i> -Obj	*	25c
LD <i>wh</i> -Subj	*	25d
<i>wh</i> -adjunct	*	26b

6. Brazilian Portuguese regular subjects occupy Spec,A/A'P

In this section I will claim that, just like locally moved *wh*-subjects, regular subjects in Brazilian Portuguese also occupy Spec,A/A'P.

6.1. Brazilian Portuguese regular subjects have mixed A/A' properties

Being in a mixed A/A' position, we expect BP subjects to exhibit both A and A' properties. Lacerda (2020) argues that BP subjects can receive an aboutness (AT) topic interpretation even without being in Spec,CP.⁷ In (28) we see that the aboutness topic object must be fronted to be licensed. *Já* (lit. "already") is an aboutness-shifting particle, used to introduce a (non-contrastive) discourse topic.

(28) A: O Pedro teve que ler o Programa Minimalista pra esse curso.

'Pedro had to read the Minimalist Program for this course.'

B1: Já o *Barriers*_{AT}, ele não teve que ler.

JÁ the *Barriers* he not had that to.read

B2: *Ele não teve que ler já o *Barriers*.

he not had that to.read JÁ the *Barriers*_{AT}

'As for *Barriers*, he didn't have to read it'

Lacerda (2020:25)

(28) shows that the aboutness topic object must be interpreted in a sentence initial position. This position, however, does not have to be Spec,CP.

⁷An anonymous reviewer pointed out that I misrepresented Lacerda's claim, "which actually states that subject in BP can be interpreted as topic even when they occupy Spec,TP". This is indeed the author's actual claim, but there is no A/A'P projection in his theoretical assumptions. If we interpret BP subjects as occupying Spec,A/A'P instead of Spec,TP we account for the topic reading without having to assume TP in the language hosts [Topic] features, which seems to treat BP as typologically exceptional. There doesn't seem to have any empirical loss in proposing BP subjects are actually in the A/A'P projection instead, which is equipped to handle both A (e.g., agreement) and A' (e.g., topic reading) properties. Additional motivation is required to sustain TP does the job, where this is expected under the A/A'P analysis.

(29) A: O Pedro leu dez livros do Chomsky pra esse curso.
 the Pedro read ten books of.the Chomsky to this course
 'Pedro read ten books by Chomsky for this course.'

B: Já o João_{AT} não leu nenhum.

JÁ the João not read.3SG none

'As for João, he didn't read any.'

Lacerda (2020:209)

The aboutness topic DP *o João* in (29) corresponds to a regular subject, thus triggering agreement in (29B). Also notice that no pause is required after the subject, contrary to B1 in (28). These facts can be captured if the subject is in Spec,A/A'P, a position that is able to license both A (e.g., agreement) and A' (e.g., topic) properties. Importantly, English does not allow subject topics (cf. Lasnik & Saito 1994, Bošković 2016).

Martins & Nunes (2010) and Rodrigues (2004) also observe that BP DP subjects exhibit topic behavior. For instance, topicalized possessives can occupy subject position, thus triggering verbal agreement. Also notice that such topic-subjects can hyper-raise (31).

(30) a. Os ponteiros do relógio quebrara-m.
 the hands of.the watch broke-PL
 'The hands of the watch broke.'

b. O relógio quebrou os ponteiros.
 the watch broke.SG the hand
 'As for the watch, its hand broke.'

(31) O relógio parece que quebrou os ponteiros.
 the watch seems that punctured the hands
 'As for the watch, it seems that its hands broke.'

The above is for the A' properties associated with A/A'P. For the A properties, we have seen that they trigger verbal agreement (cf. 2a, repeated below as 32).

(32) Os meninos parecem que *t* comeram maçã.
 the boys seem.3PL that ate.3PL apple
 'The boys seem to have eaten apples'

The data in section 2 show that HR in BP can be characterized as A-movement. For instance, we saw in (7b), repeated below as (33a), that DP subjects of sentential idioms, which generally resist A'-movement (cf. 33b), can hyper-raise. Only a non-idiomatic reading is available for (33b).

(33) a. A vaca parece que foi pro brejo.
 the cow seems that went to.the swamp
 'It seems that things went bad.'

b. *A vaca parece que ela foi pro brejo.
 the cow seems that it went to.the swamp
 (Int.) 'It seems that things went bad.'

Additionally, Fong (2018) shows HR in BP has another property of A-movement, namely, that hyper-raised subjects are able to bind variables without inducing weak cross-over effects.

- (34) Nenhum aluno₁ parece pro orientador dele₁ [t que entendeu a lição]
 no student seem.3SG for.the advisor of.he that understood the lesson
 'No student seems to his advisor to have understood the class.' Fong (2018:11)

Lastly, Ferreira observes that this position hosts negative quantifiers, which cannot be topicalized (i.e., cannot be in a pure A' position).⁸

- (35) a. Ninguém parece que entendeu o problema.
 nobody seems that understood the problem
 'Nobody seems to have understood the problem.'
 b. *Ninguém, o João disse que entendeu o problema.
 nobody the João said that understood the problem
 (Int.) 'João said that nobody understood the problem.' Nunes (2008:09)

I therefore propose that BP hyper-raising constructions are derived via movement from Spec,A/A'P in the embedded clause to Spec,A/A'P in the matrix one (cf. 1, repeated below as 36).

- (36) [CP [A/A'P DP_i [TP ... [CP [A/A'P t_i [TP ...

HR is then predicted only in languages where regular subjects can be in Spec,A/A'P. This is not improper movement, since it is not movement from a pure A' position (e.g., Spec,CP) to a pure A position (e.g., Spec,TP).

6.2. Scope rigidity and hyper-raising

We have seen that in English, even though regular DP subjects allow for an inverse scope reading (cf. 20a, repeated below as 37a), *wh*-subjects don't (cf. 20b, repeated below as 37b).

- (37) a. Someone loves everyone. ($\exists > \forall$; $\forall > \exists$)
 b. Who loves everyone. ($\exists > \forall$; $*\forall > \exists$) Bošković (2021)

⁸An anonymous reviewer acknowledged the discussion in the literature that shows the position occupied by hyper-raised subjects in BP does not count as an A'-position. They mention the fact HR subjects are compatible with idioms (38) and indefinite quantifiers (40), which cannot move to an A'-topic like position. This analysis does not state HR subjects go to a **pure** A'-position (i.e., Spec,CP), where such elements are definitely not expected. Crucially, the A/A' position is able to host elements that have performed A or A' movement. In other words, the A aspect of it is able to admit either a subject raised out of a sentential idiom or an indefinite quantifier. Stating it in terms of Boolean operators, the A/A' projection has to be understood as A **or** A', not as A **and** A', i.e., A-movement, A'-movement, and mixed A/A'-movement are all predicted to be allowed in Spec,A/A'P. This analysis doesn't imply that every movement targeting Spec,A/A'P should obligatorily exhibit mixed properties. An account that treats BP subjects as raisers to Spec,TP shouldn't expect it to exhibit A' discourse effects of the type mentioned here, since the language has a dedicated (pure) A' position for that (Spec,CP). Given there is independent motivation for Spec,A/A'P, it is to the advantage of this analysis to capitalize on it to account for subjects which display both A and A' characteristics.

Interestingly, (regular) BP subjects behave like wh-subjects regarding scope interaction. Assuming that the absence of inverted scope reading in (37b) is due to the wh-subject occupying a position higher than Spec,TP, thus blocking object inversion, I argue non-wh BP subjects also occupy this position (i.e., Spec,A/A'P). This provides a straightforward account of why BP resists an inverted scope reading in (38a). Notice that under HR the inverse scope reading is also impossible (cf. 38c).⁹

- (38) a. Alguém ama todo mundo. ($\exists > \forall$; $*\forall > \exists$)
 someone loves everyone
 b. Quem ama todo mundo? ($\exists > \forall$; $*\forall > \exists$)
 who loves everyone
 c. Alguém parece que ama todo mundo. ($\exists > \forall$; $*\forall > \exists$)
 someone seems that loves everyone

The proposal above gives rise to an empirical prediction: languages that exhibit hyper-raising are expected to be scope-rigid, at least in "subject precedes object" cases such as (38a). Notice that this is a one-way correlation: absence of inverse scope reading will not mean availability of HR. So far, the prediction is indeed borne out for several languages from different families listed in Ura (1994:65) as allowing HR. Here, I list HR languages that are also scope-rigid:

(39) Scope-rigid hyper-raising languages:

Mandarin Chinese (cf. Huang 1998), Japanese (cf. Hoji 1985), Korean (cf. Kim 2008), Persian (cf. Shafiei & Storoshenko 2017), Turkish (cf. Altinok 2017), Greek (cf. Baltazani 2002), Kinande (Bantu; cf. Baker 2003), Georgian (cf. Borise & Polinsky 2018), Finnish (cf. Zimmermann 2003), Tamil (cf. Sarma 2003), Sinhala (cf. Tim Chou & Hettiarachchi 2016). Hungarian, which exhibits HR (cf. den Dikken 2017), is also scope-rigid (cf. Kiss 2002). Spanish, which has HR according to Fernández-Salgueiro 2005 and Pires & Nediger 2018), is also claimed to be scope-rigid (cf. Alexiadou & Anagnostopoulou 1998).

7. Deriving hyper-raising in Brazilian Portuguese: the voiding of freezing effects

According to Bošković (2008), once X enters a feature checking relation, it is frozen in place and can no longer move. Crucially, X can only be probed once, regardless of the status of the probe (i.e., A or A').

- (40) X undergoes A [or] A' movement feature checking as a probe only once. Bošković (2008:24)

So:

⁹An anonymous reviewer said the data in (38) are not helpful because "Brazilian Portuguese displays surface scope even when subjects are not involved" as in *O João apresentou alguém pra todo mundo* ('João introduced someone to everyone'), with the existential quantifier obligatorily scoping over the universal quantifier. I don't see how this sentence is relevant in the context of the present discussion. The claim here is that a quantifier in object position cannot scope over the quantifier in subject position. The fact a direct object takes scope over an indirect object in a ditransitive construction tells us nothing about the unavailability of inverse scope in (38). I have never claimed the A/A' position accounts for all the scope-rigid readings of the language.

A: *[A'] feature checking movement feeding [A'] feature checking movement. Cf. Rizzi's (2006,2007) Criterial Freezing.¹⁰

B: *[A] feature checking movement feeding [A'] feature checking movement. Cf. *who left?* effect: *wh*-subject movement cannot proceed through Spec,TP (Bošković 2019, Messick 2020).¹¹

We also do not expect C or D:

C: *[A] feature checking movement feeding [A] feature checking movement. Cf. Chomsky's (2000,2001) Activation Condition.

D: *[A'] feature checking movement feeding [A] feature checking movement; i.e., improper movement.¹²

I therefore propose (41):

(41) No freezing effect is expected where the A/A' projection is involved, since in this case we don't have A, B, C, or D.

With this approach, hyper-raising is a side effect of subjects occupying Spec,A/A'P, a position where licensing of both A and A'-features are possible. The mixed A/A' position is then the escape hatch needed to void freezing effects. The derivation of the HR sentence *Os meninos parecem que comeram maçã* (cf. 32) is given below.

(42) [_{CP} [_{A/AP} *Os meninos*]_i [_{TP} *parecem* [_{VP} *t_i* [_{CP} *que* [_{A/AP} *t_i* [_{TP} *comeram* [_{VP} *t_i* *maçã* ...

The A/A' projection is always involved in the derivation above: the subject DP moves from Spec,vP to Spec,A/A'P in the embedded clause. Then, it moves to the matrix Spec,vP, and finally to matrix Spec,A/A'P. Therefore, in this approach HR is a side effect of regular subjects occupying Spec,A/A'P.

8. Conclusion

In this paper I have argued that the availability of hyper-raising in Brazilian Portuguese is associated with its subject occupying Spec,A/A'P, a mixed A/A' position between CP and TP in the left periphery of clause structure dedicated to locally moved subjects, thus accounting for the mixed A/A' properties of Brazilian Portuguese subjects. I have also claimed that the freezing effect of feature-checking is voided only when A/A' movement is involved.

¹⁰No A'-movement (e.g., *wh*-movement, topicalization, focalization, QR, NPI-licensing) can feed another instance of A'-movement.

¹¹This approach also captures Chomsky's Activation Condition if interpreted as [A] feature checking movement feeding [A] feature checking movement.

¹² No Spec,TP to Spec,CP movement in *who left?*. Also, in Kinande, objects agree with the verb only when there is no *wh*-movement of the object. So, it is not possible to do object shift (i.e., movement to Spec,vP) followed by *wh*-movement (i.e., movement to Spec,CP), given D (Bošković, p.c.).

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